Xi'an Jiaotong-Liverpool University

Xi'an Jiaotong-Liverpool University 西交利物浦大學

PAPER CODE	EXAMINERS	DEPARTMENT	TEL
CPT 107	K.L. Man and Gabriela Mogos	CPT	1509

2022/23 SEMESTER 1 – TAKE-HOME OPEN BOOK RESIT EXAM

BACHELOR DEGREE – YEAR 2

DISCRETE MATHEMATICS AND STATISTICS

DURATION: 2 HOURS

CRASH TIME: 15 MINS

INSTRUCTIONS TO CANDIDATES

1. The resit exam is a take-home open book exam and the work should be done independently.
2. Total marks available are 100.
3. The number in the column on the right indicates the marks for each question.
4. Answer all questions.
5. Answers should be written in English.
6. Relevant and clear steps should be included in your answers.
7. Your answers should be submitted electronically through the Learning Mall via the submission link.
8. The naming of Report (in pdf) is as follows: CPT107_StudentID_RESIT (e.g., CPT107_12345678_RESIT.pdf).
9. Answers can also be handwritten, fully and clearly scanned or photographed for submission as one single PDF document through the Learning Mall via the submission link.

Notes:

- **To obtain full marks for each question, relevant and clear steps need to be included in the answers.**
- **Partial marks may be awarded depending on the degree of completeness and clarity.**

Question 1: Proof Techniques **[7 marks]**

(a) For $r, s \in \mathbb{Z}$, if the product of *r* and *s* is even, then *r* is even or *s* is even. Prove or disprove it.

(4 marks)

(b) For all integers $m \geq 1$, use proof by induction to show that:

$$2 \cdot 2^1 + 3 \cdot 2^2 + 4 \cdot 2^3 + \cdots + (m+1) \cdot 2^m = m \cdot 2^{m+1}.$$

(3 marks)

Question 2: Set Theory **[12 marks]**

(a) Let the universal set be $\mathbb{Z}$, $\mathrm{R} = \{\mathrm{x} \in \mathbb{Z} \mid \mathrm{x} \text{ is odd and } 2 < \mathrm{x} < 11\}$, $\mathrm{S} = \{\mathrm{x} \in \mathbb{Z} \mid \mathrm{x} \text{ is even and } 1 < \mathrm{x} < 8\}$ and $\mathrm{T} = \{\mathrm{x} \in \mathbb{Z} \mid \mathrm{x} \text{ is odd and } 2 < \mathrm{x} < 9\}$. Find the elements of:

1. $\mathrm{R} \cap (\mathrm{S} \cup \mathrm{T})$
2. $\sim((\mathrm{R} - \mathrm{T}) \cap \mathrm{S})$
3. $(\mathrm{T} \cup \mathrm{R}) \cap (\mathrm{R} \,\Delta\, \mathrm{S})$
4. $\big((\mathrm{T} \cup \mathrm{R}) \cap (\mathrm{R} \,\Delta\, \mathrm{S})\big) - \big(\mathrm{R} \cap (\mathrm{S} \cup \mathrm{T})\big)$

(4 marks)

(b) Let *A*, *B*, *C* and *D* be sets. If $A \subset B$, $B \subset C$ and $C \subset D$, then $A \subset D$. If you think that it is true, prove it. If not, explain why.

(5 marks)

(c) Let *A* be a set. Prove or disprove that $A \subseteq Pow(A)$.

(3 marks)

Question 3: Relations **[21 marks]**

(a) Let $N = \{4,1,3,2\}$. What is the transitive closure of the relation: $\{(4,3),(4,2),(1,4),(3,1)\}$ on N?

(4 marks)

(b) Let x and y be integers, and R be the relation on the set of integers given by xRy such that $2x=1+y$. Prove or disprove that R is both reflexive and symmetric, and R is an equivalence relation.

(5 marks)

(c) Let R be a relation. Prove or disprove that if $R^2 \subseteq R$ then R is transitive.

(6 marks)

(d) Let $X = \{a, b, c, d, e\}$ and let S be a relation on X such that: $S = \{(a,a), (b,b), (c,c), (d,d), (e,e), (a,b), (a,c), (a,d), (a,e), (d,e)\}$. Prove or disprove that S is a partial order. If S is a partial order, draw its Hasse diagram and further prove or disprove that S is a total order.

(6 marks)

Question 4: Functions and PHP **[17 marks]**

(a) Let A be the set of even integers and let B be the set of odd integers.
Consider $f: A \to B$ defined by $f(x) = x + 1$.

1. Prove that f is injective. (3 marks)
2. Prove that f is surjective. (3 marks)
3. Find the inverse of f. (3 marks)

(b) If f(x) and g(x) are both even functions, prove or disprove that $(g \circ f)(x)$ is an even function.

(5 marks)

(c) Find the minimum number of elements that one needs to take from the set S = {1, 2, 3, . . ., 9} to be sure that two of the numbers are added up to 10.

(3 marks)

Question 5: Logic **[24 marks]**

(a) Show whether the statements below are tautologies and/or contradictions:

1. $(\neg p \wedge (p \vee q)) \rightarrow q$ (2 marks)
2. $((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$ (2 marks)
3. $\neg((p \wedge \neg p) \rightarrow q)$ (2 marks)

(b) For each of the two cases 1. and 2. below, show whether the statement on the left-hand side is equivalent to the statement on the right-hand side:

1. $p \wedge (q \vee r \vee s) \equiv (p \wedge q) \vee (p \wedge r) \vee (p \wedge s)$ (3 marks)
2. $p \wedge (\neg(q \wedge r)) \equiv (p \wedge \neg q) \vee (p \wedge \neg r)$ (3 marks)

(c) Consider the signature S = *{Friend, Male, john, rebecca}* consisting of a binary predicate symbol *Friend,* a unary predicate symbol *Male*, and two constant symbols *john* and *rebecca*. Assume that these symbols have the following meaning:

- *Friend* means "is a friend of" (i.e., *Friend (x, y)* states x is a friend of y).
- *Male* means "is male" (i.e., *Male(x)* states x is male).
- *john* and *rebecca* refer to "John" and "Rebecca", respectively.

Translate the following sentences into S-formulae, that is, for each of the following sentences provide an S-formula that expresses the sentences:

1. Rebecca is a friend of John.
2. John has a male friend.
3. All friends of John are also friends of Rebecca.
4. John has at least 2 friends.
5. Everybody has a friend.
6. Nobody is a friend of everybody else.

(12 marks)

Question 6: Combinatorics and Probability **[19 marks]**

(a) In how many ways, 8 people can sit in a row if:

1. there are no restrictions on the seating arrangements.
2. eight people are 4 men and 4 women, and no 2 men or 2 women can sit next to each other.
3. eight people are 5 men and 3 women, and 5 men must sit next to each other.
4. eight people are 4 married couples, and each couple must sit together.

(8 marks)

(b) A box contains 2 red balls, 3 green balls, and 5 blue balls. A ball is selected at random, and its color is noted. Then it is replaced, and another ball is selected, and its color is noted. Find the probability of each of these:

1. Selecting 2 blue balls (2 marks)
2. Selecting a blue ball and then a red ball (2 marks)
3. Selecting a green ball and then a blue ball (2 marks)

(c) John tosses two fair coins. He wins \$5 if 2 heads occur, \$2 if 1 head occurs, and \$1 if no head occurs.

1. Calculate the expected value of his game. (4 marks)
2. How much should John pay to play the game if it is to be fair? (1 mark)

END OF RESIT PAPER